# Project Title: ComicCraft – AI Comic Story Creator Using Gemini Models

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## 2. Project Overview

ComicCraft is an AI-powered application that helps users create comic stories from simple ideas. The user provides a story concept, characters, genre, tone, setting and desired number of panels. A Gemini model generates a structured comic story containing a title, storyline, scenes, actions, dialogue and narration. The application presents the result in a clear panel-by-panel format.

## 3. Problem Statement

Creating a complete comic manually requires planning characters, scenes, dialogue and story progression. Beginners may find it difficult to convert an idea into a consistent panel sequence. ComicCraft reduces this effort by using generative AI to transform user ideas into an editable comic story.

## 4. Objectives

- Build an easy-to-use AI comic story creator.
- Integrate Gemini models for story generation.
- Convert a user idea into a structured panel-by-panel comic.
- Allow customization of genre, tone, characters, setting and panel count.
- Provide edit and regenerate options.
- Validate inputs and handle AI/API errors.
- Keep Gemini credentials secure.
- Provide optional project saving and export functionality.

## 5. Scope

| Area | Scope |
|---|---|
| Input | Story idea, genre, tone, characters, setting, panel count |
| AI | Gemini-based story generation |
| Output | Title, summary and comic panels |
| Editing | Modify generated text and regenerate |
| Storage | Optional project history/database |
| Export | Optional PDF/text/image export |

## 6. Target Users

- Students and project learners.
- Comic writers and creative users.
- Teachers demonstrating generative AI.
- Developers learning Gemini API integration.

## 7. Functional Requirements

| ID | Requirement | Description |
|---|---|---|
| FR-01 | Story Idea | User can enter a comic idea. |
| FR-02 | Genre | User can select a genre. |
| FR-03 | Characters | User can enter character names and descriptions. |
| FR-04 | Setting | User can specify the story world/location. |
| FR-05 | Tone | User can select a tone such as funny, dramatic or suspenseful. |
| FR-06 | Panel Count | User can choose a valid number of panels. |
| FR-07 | Generate | System sends validated input to Gemini and receives generated content. |
| FR-08 | Display | System displays the result as comic panels. |
| FR-09 | Regenerate | User can request an alternative result. |
| FR-10 | Edit | User can edit generated content. |
| FR-11 | Save | System may save projects. |
| FR-12 | Export | System may export the final story. |
| FR-13 | Errors | System displays useful validation/API error messages. |

## 8. Non-Functional Requirements

| Category | Requirement |
|---|---|
| Performance | Show loading state during AI generation and avoid duplicate requests. |
| Security | Keep Gemini credentials on the server side. |
| Usability | Provide a simple and readable interface. |
| Reliability | Handle API, network and timeout failures |

| | |
|---|---|
| | gracefully. |
| Scalability | Allow additional AI features later. |
| Maintainability | Separate frontend, backend and AI integration logically. |
| Accessibility | Use readable labels, buttons and keyboard-friendly controls. |
| Privacy | Store only necessary user/project information. |

## 9. Suggested Technology Stack

| Layer | Suggested Technology |
|---|---|
| Frontend | HTML, CSS, JavaScript / React / Next.js |
| Backend | Node.js + Express or Next.js API routes |
| AI | Google Gemini API / Gemini models |
| Database | Firebase/Firestore, PostgreSQL or MongoDB (optional) |
| Authentication | Firebase Authentication or equivalent (optional) |
| Version Control | Git and GitHub |
| Hosting | Suitable cloud platform |

## 10. System Architecture

Recommended flow: User → Frontend Form → Backend/API → Gemini Model → Response Validation/Parsing → Comic Viewer → Optional Database/Export.

| Component | Responsibility |
|---|---|
| Frontend | Collect input and display comic panels. |
| Backend/API | Validate input, construct prompt, call Gemini and return structured output. |
| Gemini Model | Generate the requested comic story. |
| Database | Store users/projects when persistence is enabled. |
| Export | Convert final content to supported format. |

## 11. Core Modules

| Module | Description |
|---|---|
| Home | Project introduction and start button. |
| Story Creation | Collect idea, genre, characters, setting, tone and panel count. |
| Gemini Integration | Securely send requests to Gemini and receive results. |
| Story Parser | Convert AI response into expected panel structure. |
| Comic Viewer | Display title and panels. |
| Editor | Edit and regenerate content. |
| History | Save and reopen projects if implemented. |
| Export | Export the completed comic if implemented. |
| Validation/Error | Handle invalid input and service failures. |

## 12. User Interface Requirements

- Landing page with ComicCraft title and Create Comic button.
- Clearly labelled story creation form.

- Genre and tone selection.
- Character input section.
- Setting input.
- Panel-count selector.
- Generate button with loading state.
- Panel-by-panel generated output.
- Edit and Regenerate controls.
- Save/Export controls where implemented.
- Responsive design for supported devices.

## 13. Input Requirements

| Input | Example | Validation |
|---|---|---|
| Story Idea | A student discovers a robot at college | Required; length limit |
| Genre | Science Fiction | Required selection |
| Tone | Funny | Optional/default |
| Characters | Arun – student; Robo – friendly robot | At least one recommended |
| Setting | Modern college campus | Optional/required by design |
| Panel Count | 6 | Valid integer within configured range |
| Language | English | Optional feature |

## 14. Gemini Prompt Design

Use a clear prompt that describes the user's requirements and the exact output structure. For reliable application parsing, request structured output such as JSON when supported by the selected Gemini API/model configuration.

- title
- summary
- characters
- panels
- scene_description
- action
- dialogue
- narration
- ending

## 15. Example Prompt Template

```
Create a comic story based on these requirements:
Story idea: {storyIdea}
Genre: {genre}
Tone: {tone}
Setting: {setting}
Characters: {characters}
Number of panels: {panelCount}

Create a coherent beginning, middle and ending.
```

```
For every panel provide:
- panel number
- scene description
- character actions
- dialogue
- narration
Keep character names consistent.
Return the result in the structured format required by the application.
```

## 16. Gemini API Integration Requirements

- Configure Gemini access according to the current Google AI/Gemini setup used by the project.
- Store API credentials in environment variables or a secure secret manager.
- Never place the secret API key in public frontend code.
- Create a backend endpoint such as /api/generate-comic.
- Validate request data before calling Gemini.
- Build the prompt from validated values.
- Call the selected Gemini model using the supported SDK/API.
- Handle authentication, rate-limit, timeout and service errors.
- Parse/validate the model response before sending it to the frontend.

## 17. Backend API Requirements

| Endpoint | Method | Purpose |
|---|---|---|
| /api/generate-comic | POST | Generate comic using Gemini. |
| /api/projects | GET | List saved projects (optional). |
| /api/projects | POST | Save a project (optional). |
| /api/projects/:id | GET | Open a saved project (optional). |
| /api/projects/:id | PUT | Update project (optional). |
| /api/projects/:id | DELETE | Delete project (optional). |

## 18. Database Requirements

| Entity | Suggested Fields |
|---|---|
| User | id, name, email, createdAt |
| Project | id, userId, title, promptData, generatedStory, createdAt, updatedAt |
| Character | id, projectId, name, role, description |
| Panel | id, projectId, panelNumber, scene, action, dialogue, narration |

A simple implementation can store the structured story as one JSON document. A normalized database is useful when characters and panels must be edited or searched independently.

## 19. Step-by-Step Development Plan

| Phase | Activities |
|---|---|
| 1. Planning | Finalize requirements, user flow, wireframe and data model. |
| 2. Frontend | Build landing page and story form. |
| 3. Backend | Create API endpoint and validation. |
| 4. Gemini Integration | Connect Gemini and test prompts. |

| | |
|---|---|
| 5. Output | Parse and display structured panels. |
| 6. Editing | Add edit/regenerate controls. |
| 7. Persistence | Add database/history if required. |
| 8. Export | Implement supported export. |
| 9. Testing | Perform functional, UI, security, error and AI-output tests. |
| 10. Deployment | Configure environment variables and deploy. |

## 20. Testing Requirements

| Test ID | Test Case | Expected Result |
|---|---|---|
| TC01 | Valid story input | Comic generated successfully. |
| TC02 | Missing story idea | Validation message; no invalid request. |
| TC03 | Invalid panel count | Invalid value rejected. |
| TC04 | Gemini/API failure | Useful error shown and retry possible. |
| TC05 | Regenerate | Alternative valid result generated. |
| TC06 | Character consistency | Names remain consistent across panels. |
| TC07 | Long input | Input handled within limits. |
| TC08 | API key exposure check | Secret is not present in client code. |
| TC09 | Save project | Project can be stored/reopened if implemented. |
| TC10 | Responsive UI | Application remains usable on supported screens. |

## 21. AI Safety and Responsible Use

- Validate user input and generated output.
- Protect API credentials.
- Use safety controls appropriate to the chosen AI platform and target audience.
- Avoid storing unnecessary personal information.
- Tell users that AI output may contain mistakes and should be reviewed.
- Provide edit/regenerate controls for unsuitable results.

## 22. Security Requirements

- Use environment variables/secret manager for credentials.
- Validate and sanitize API requests.
- Apply rate limiting where appropriate.
- Protect private project endpoints with authentication/authorization.
- Do not trust client-supplied user IDs for authorization.
- Use HTTPS in production.
- Never log API keys or sensitive secrets.

## 23. Performance Requirements

- Show a loading indicator while generating.

- Disable repeated Generate clicks during an active request.
- Set reasonable input and panel limits.
- Use timeout/error handling.
- Avoid unnecessarily sending repeated large prompts.

## 24. Expected Output

A successful generation should display a comic title, short summary and a sequence of panels. Each panel should contain a scene description, character actions, dialogue and/or narration, with a logical story progression from beginning to ending.

## 25. Suggested Screenshot Captions

Figure 1: ComicCraft Home Page

Figure 2: Story Creation Form

Figure 3: Genre and Tone Selection

Figure 4: Character Details Input

Figure 5: Gemini API Integration Configuration

Figure 6: Comic Generation Request

Figure 7: Generated Comic Story

Figure 8: Generated Comic Panels

Figure 9: Comic Story Editing Screen

Figure 10: Regenerate Comic Option

Figure 11: Saved Comic Project

Figure 12: Project History Screen

Figure 13: Export Comic Option

Figure 14: Error Handling / Validation Message

Figure 15: Final ComicCraft Output

## 26. Project Deliverables

- Complete source code.
- Frontend interface.
- Backend/API integration.
- Gemini integration.
- Prompt templates and structured output format.
- Database configuration if used.
- Test cases and results.

- Screenshots with captions.
- Project report.
- README with setup and execution instructions.

## 27. Final Checklist

- Requirements finalized.
- Story creation form completed.
- Gemini integration completed.
- API key secured.
- Input validation implemented.
- Structured AI output implemented.
- Comic viewer completed.
- Edit/regenerate tested.
- Database/history completed if required.
- Error handling tested.
- Security checks completed.
- Test cases documented.
- Screenshots captured and captioned.
- README and report prepared.

## 28. Conclusion

ComicCraft demonstrates how Gemini-based generative AI can be integrated into a practical creative web application. By combining a simple interface, secure backend integration, structured AI output, panel-based presentation and testing, the project provides a complete foundation for an academic generative-AI application.

## 29. Viva Questions and Answers

**Q: What is ComicCraft?**

A: An AI-powered application that generates comic stories using Gemini models.

**Q: Why is Gemini used?**

A: It provides generative AI capabilities for creating story content from user prompts.

**Q: What does the backend do?**

A: It validates input, securely calls Gemini and returns structured output.

**Q: Why should the API key be hidden?**

A: To prevent unauthorized users from obtaining and misusing the credential.

**Q: What is structured output?**

A: A predictable format containing fields such as title, characters and panels.

**Q: What is regeneration?**

A: Sending another generation request to create an alternative result.

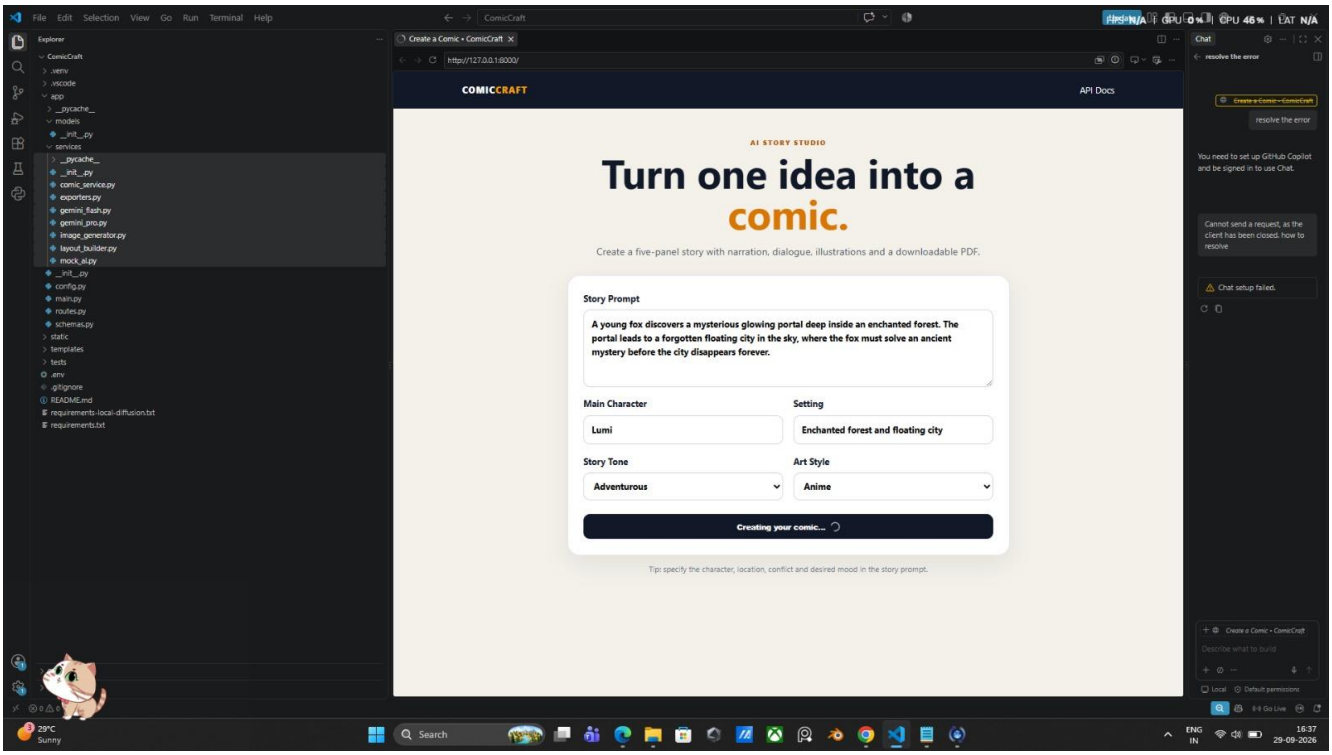
**Q: Why is validation required?**

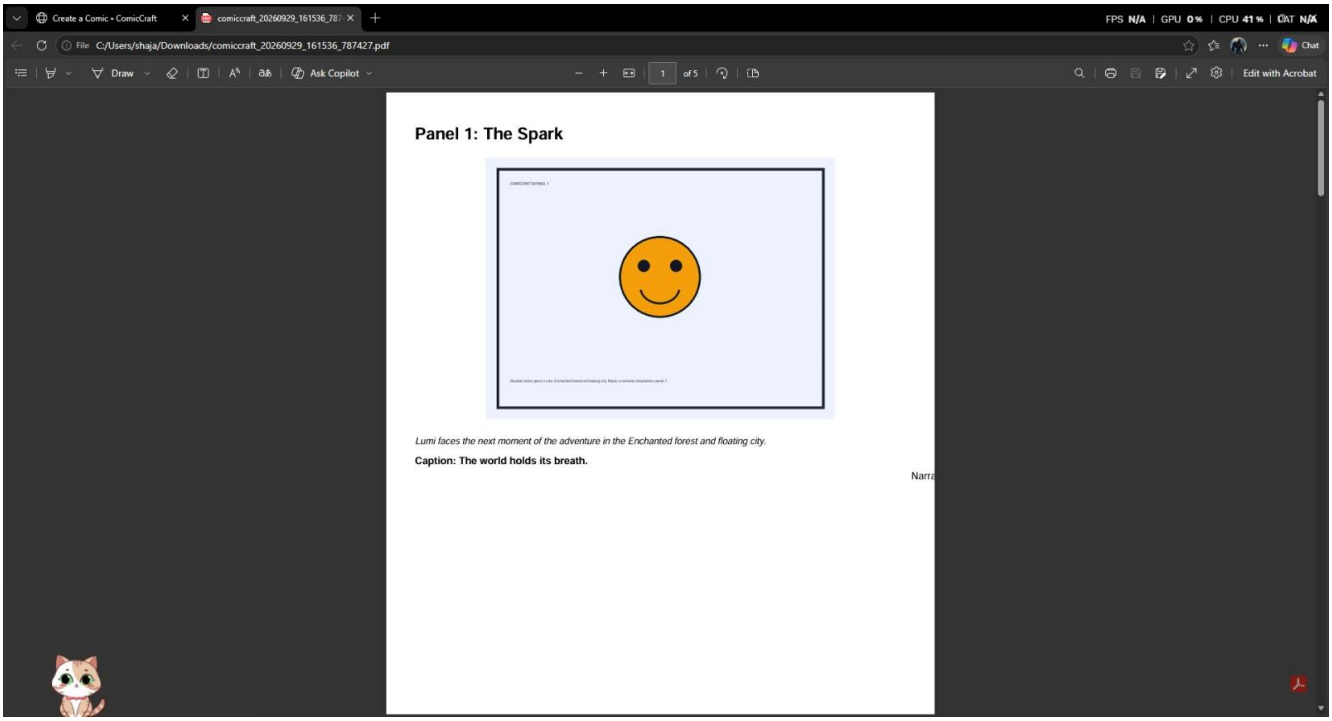
A: To prevent invalid requests and improve reliability.

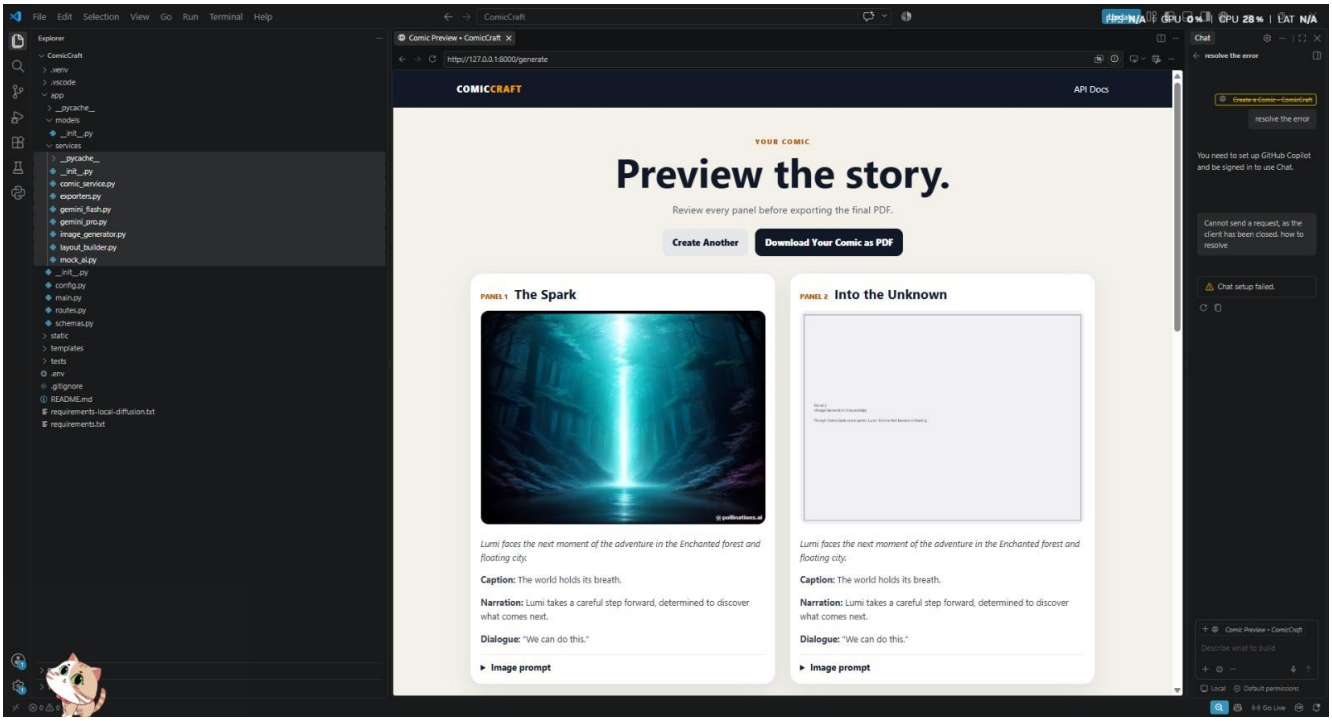
**Q: What are the main modules?**

A: Story creation, Gemini integration, parser, comic viewer, editor, history and export.

# Screen Shot:




Create a Comic • ComicCraft
comiccraft_20260929_161536_787
File C:/Users/shaja/Downloads/comiccraft_20260929_161536_787427.pdf
Draw
Ask Copilot
1 of 5
Edit with Acrobat
Panel 1: The Spark
Lumi faces the next moment of the adventure in the Enchanted forest and floating city.
Caption: The world holds its breath.


COMICCRAFT
API Docs
YOUR COMIC
Preview the story.
Review every panel before exporting the final PDF.
Create Another
Download Your Comic as PDF
PANEL 1 The Spark
PANEL 2 Into the Unknown
Lumi faces the next moment of the adventure in the Enchanted forest and floating city.
Caption: The world holds its breath.
Narration: Lumi takes a careful step forward, determined to discover what comes next.
Dialogue: "We can do this."
Image prompt

comiccraft_20260929_163758_674 ×

FPS N/A | GPU 0% | CPU 30% | NPU N/A

File C:/Users/shaja/Downloads/comiccraft_20260929_163758_674426.pdf

Chat

Draw | Ask Copilot | 1 of 5 | Edit with Acrobat

## Panel 1: The Spark




*Lumi faces the next moment of the adventure in the Enchanted forest and floating city.*

**Caption: The world holds its breath.**

Narra



Create a Comic • ComicCraft ×

FPS N/A | GPU 0% | CPU 48% | NPU N/A

127.0.0.1:8000

Chat

AI STORY STUDIO

# Turn one idea into a comic.

Create a five-panel story with narration, dialogue, illustrations and a downloadable PDF.

**Story Prompt**

A young fox discovers a mysterious glowing portal deep inside an enchanted forest. The portal leads to a forgotten floating city in the sky, where the fox must solve an ancient mystery before the city disappears forever.

**Main Character**

Lumi

**Setting**

Enchanted forest and floating city

**Story Tone**

Light-hearted

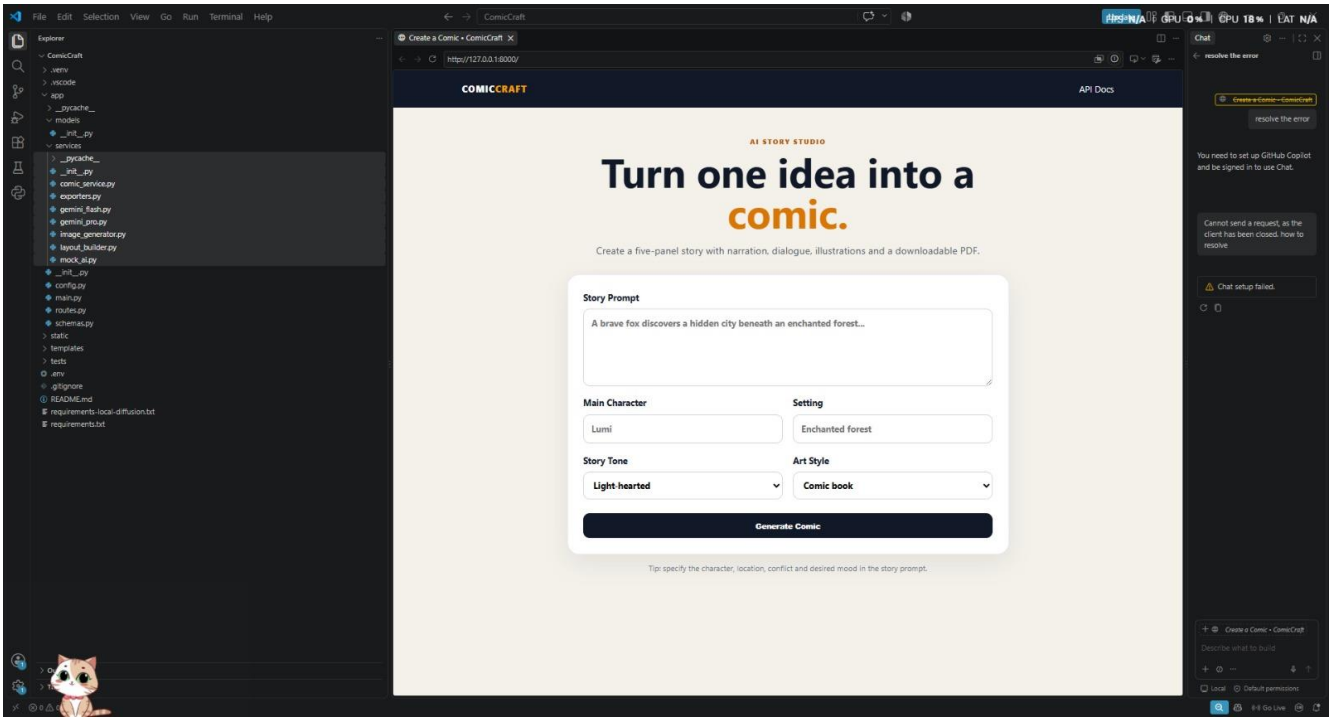
**Art Style**

Comic book

Creating your comic...

Tip: specify the character, location, conflict and desired mood in the story prompt.




File Edit Selection View Go Run Terminal Help
ComicCraft
Explorer
ComicCraft
.venv
.vscode
app
__pycache__
models
__init__.py
services
__pycache__
__init__.py
comic_service.py
exporters.py
gemini_flash.py
gemini_pro.py
image_generator.py
layout_builder.py
mock_ai.py
__init__.py
config.py
main.py
routes.py
schemas.py
static
templates
tests
.env
.gitignore
README.md
requirements-local-diffusion.txt
requirements.txt
Create a Comic • ComicCraft
http://127.0.0.1:8000/
COMICCRAFT
API Docs
AI STORY STUDIO
Turn one idea into a comic.
Create a five-panel story with narration, dialogue, illustrations and a downloadable PDF.
Story Prompt
A brave fox discovers a hidden city beneath an enchanted forest...
Main Character
Lumi
Setting
Enchanted forest
Story Tone
Light-hearted
Art Style
Comic book
Generate Comic
Tip: specify the character, location, conflict and desired mood in the story prompt.
Chat
resolve the error
Create a Comic • ComicCraft
resolve the error
You need to set up GitHub Copilot and be signed in to use Chat.
Cannot send a request, as the client has been closed. how to resolve
Chat setup failed.
Create a Comic • ComicCraft
Describe what to build
Local
Default permissions
Go Live